CHAPTER 3

CURRENT ELECTRICITY

CHAPTER ESSENCE AT A GLANCE:
The microscopic dynamics of charge drift velocity, Ohm's law, circuit analysis with Kirchhoff's laws, and Wheatstone bridge networks.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Drift Velocity & Electric Current:** Conduction electrons acquire a steady net drift velocity directed opposite to the applied electric field, constituting macroscopic electric current.

• **Ohm's Law & Resistivity:** Current density remains proportional to applied electric field in ohmic conductors; electrical resistance is directly proportional to length and resistivity, and inversely proportional to cross-sectional area.

• **Electromotive Force & Internal Resistance:** Maximum potential difference across open cell terminals; during current discharge, internal resistance produces a terminal voltage drop proportional to the drawn current.

• **Kirchhoff's Conservation Rules:** Junction rule embodies the principle of conservation of electric charge; closed loop rule expresses the principle of conservation of energy around any closed mesh.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$I = neAv_d \quad \text{and} \quad v_d = \frac{eE\tau}{m}$$

Electric current and electron drift velocity

$$R = \rho\frac{l}{A} \quad \text{and} \quad \rho = \frac{m}{ne^2\tau}$$

Resistance and resistivity formulation

$$\sum I_{\text{junction}} = 0 \quad \text{and} \quad \sum \Delta V = 0$$

Kirchhoff's Current (KCL) and Voltage (KVL) Laws

$$\frac{P}{Q} = \frac{R}{S}$$

Balanced Wheatstone's bridge condition

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never mix up the physical laws: KCL is based on Charge Conservation; KVL is based on Energy Conservation.

• **Alert:** [Valuation Rubric] Draw complete 4-resistor Wheatstone quadrilateral with galvanometer and cell .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Drift velocity and relation with electric current I = n A e v d [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Drift Velocity Definition]** Average velocity with which free conduction electrons get drifted towards the positive terminal of a conductor under the influence of an applied external electric field. Formula: v d = - (e E divided by m) τ. [1 Mark]

• **[Step 2: Derivation of I = n A e v d]** Conductor of length L, cross-sectional area A, electron number density n. Total number of free electrons in length L = n · (A L). Total charge Q = n A L e. Time taken by electrons to drift through length L: t = L divided by v d. Current I = Q divided by t = (n A L e) divided by (L divided by v d) = n A e v d. Current density J = I divided by A = n e v d. [2 Marks]

• **Key Exam Result:** I = n A e v d and J = n e v d

2. Deduction of Ohm's law and resistivity in terms of relaxation time [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Substitution]** We have I = n A e v d. Substituting drift velocity v d = (e E divided by m) τ: I = n A e [(e E divided by m) τ] = (n A e^2 τ divided by m) E. [1 Mark]

• **[Step 2: Linking to Potential V]** For conductor of length L, E = V divided by L. I = (n A e^2 τ divided by m) · (V divided by L) implies V = [m divided by (n e^2 τ)] · (L divided by A) · I. At constant temperature, all terms in bracket are constant for a given material. Thus V = R I, where R = ρ L divided by A, proving Ohm's Law. Resistivity ρ = m divided by (n e^2 τ). [2 Marks]

• **Key Exam Result:** Ohm's Law V = IR proved; Resistivity ρ = m divided by (n e^2 τ)

3. V-I characteristics of ohmic and non-ohmic conductors [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Ohmic Conductors]** Conductors that strictly obey Ohm's Law (V proportional to I). V-I graph is a straight line passing through the origin; resistance R = ΔV divided by ΔI is constant. Examples: Copper wire, metallic resistors, nichrome. [1 Mark]

• **[Step 2: Non-Ohmic Conductors]** Conductors where V-I relationship is non-linear and dynamic resistance r = dV divided by dI varies with current/voltage. Examples & graphs: 1. Semiconductor diode (exponential rise in forward bias, breakdown in reverse bias). 2. GaAs (displays negative resistance region). 3. Electric torch bulb (curve bends due to heating). [1 Mark]

• **Key Exam Result:** Ohmic: Linear V-I graph (metals) | Non-Ohmic: Non-linear V-I curve (diode, GaAs)

4. Temperature dependence of resistance and temperature coefficient [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Temperature Coefficient Formula]** Resistance at temperature T: R T = R_0 (1 + α ΔT), where α is the temperature coefficient of resistance: α = (R T - R_0) divided by (R_0 ΔT) (Unit: K^{-1} or $°C^{-1}$). [1 Mark]

• **[Step 2: Calculation]** Given resistance at 27 °C is R_1 = 100 Ω, resistance at temperature T is R_2 = 117 Ω, α = 1.70 × 10^{-4} $°C^{-1}$. ΔR = R_2 - R_1 = 117 - 100 = 17 Ω. ΔT = ΔR divided by (R_1 α) = 17 divided by (100 × 1.70 × 10^{-4}) = 17 divided by (1.70 × 10^{-2}) = 1000 °C. Temperature T = T_1 + ΔT = 27 °C + 1000 °C = 1027 °C (or 1300 K). [1 Mark]

• **Key Exam Result:** Temperature T = 1027 °C (1300 K)

5. Temperature dependence of resistance and temperature coefficient [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Temperature Coefficient Formula]** Resistance at temperature T: R T = R_0 (1 + α ΔT), where α is the temperature coefficient of resistance: α = (R T - R_0) divided by (R_0 ΔT) (Unit: K^{-1} or $°C^{-1}$). [1 Mark]

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• **Key Exam Result:** Temperature T = 1027 °C (1300 K)

6. Cells in series and parallel: equivalent EMF and internal resistance [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Cells in Series]** Two cells (E_1, r_1) and (E_2, r_2) in series: $V = V_1 + V_2 = (E_1 - I r_1) + (E_2 - I r_2) = (E_1 + E_2) - I (r_1 + r_2)$. Comparing with $V = E\ eq - I\ r\ eq$: $E\ eq = E_1 + E_2$ and $r\ eq = r_1 + r_2$. [1.5 Marks]

• **[Cells in Parallel]** Two cells (E_1, r_1) and (E_2, r_2) in parallel: $I = I_1 + I_2 = (E_1 - V)$ divided by $r_1 + (E_2 - V)$ divided by $r_2 = (E_1$ divided by $r_1 + E_2$ divided by $r_2) - V$ (1 divided by $r_1 + 1$ divided by r_2). Rearranging gives: $V = [\ (E_1$ divided by $r_1 + E_2$ divided by $r_2)$ divided by (1 divided by $r_1 + 1$ divided by $r_2)\ ] - I\ [$ 1 divided by (1 divided by $r_1 + 1$ divided by $r_2)\ ]$. $E\ eq$ divided by $r\ eq = E_1$ divided by $r_1 + E_2$ divided by r_2 and 1 divided by $r\ eq = 1$ divided by $r_1 + 1$ divided by r_2. [1.5 Marks]

• **Key Exam Result:** Series: $E\ eq = E_1 + E_2$, $r\ eq = r_1 + r_2$ | Parallel: $E\ eq = (E_1 r_2 + E_2 r_1)$ divided by $(r_1 + r_2)$

7. Electrical energy and power, heating effect [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Power & Heating Formulas]** Electrical energy $W = V\ I\ t = I^2\ R\ t = (V^2$ divided by $R)\ t$. Power $P = W$ divided by $t = V\ I = I^2\ R = V^2$ divided by R. [1 Mark]

• **[Step 2: Calculation]** Two bulbs rated (100 W, 220 V) and (60 W, 220 V) connected in series across 220 V supply. Resistances: $R_1 = V^2$ divided by $P_1 = (220)^2$ divided by 100 = 484 Ω; $R_2 = V^2$ divided by $P_2 = (220)^2$ divided by 60 = 806.7 Ω. Total series resistance $R\ s = 484 + 806.7 = 1290.7\ \Omega$. Current $I = 220$ divided by 1290.7 = 0.170 A. Power in 60 W bulb = $I^2\ R_2 = (0.170)^2 \times 806.7 = 23.4$ W. Power in 100 W bulb = $I^2\ R_1 = (0.170)^2 \times 484 = 14.1$ W. The 60 W bulb glows brighter in series! [1 Mark]

• **Key Exam Result:** In series, bulb with lower power rating (higher resistance) glows brighter